

Automated Reinforcement Learning-Driven and Segmentation-Guided Approach for Pneumothorax Detection and Diagnosis in Chest X-Rays

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Abstract—Recent progress in automated interpretation of chest radiographs has largely concentrated on constructing multi-pathology systems designed to identify diverse thoracic conditions using large-scale imaging corpora. Current Vision-Language Models (VLMs) for report generation predominantly depend on global image representations and frequently underutilize the detailed structural cues provided by medical segmentation. To overcome these constraints, we introduce a segmentation-informed reinforcement learning (RL)-based framework for pneumothorax identification and diagnosis on chest radiographs. In particular, reinforcement learning is utilized to refine the decision optimization process, allowing the model to acquire clinically relevant strategies through reward-guided signals. In addition, pneumothorax segmentation masks are integrated to highlight lesion regions and provide accurate spatial supervision, thus improving the fidelity of subsequent report generation. Empirical evaluations indicate that the proposed method more effectively aligns visual perception with textual reasoning for pneumothorax diagnosis. By combining segmentation-based guidance with reinforcement learning optimization, the VLM achieves precise location of the lesion while maintaining anatomical coherence, ultimately enhancing both model interpretability and diagnostic reliability. Comprehensive metric-based assessments demonstrate that the proposed approach effectively narrows the disconnect between coarse visual localization and higher-level textual inference derived from reports.

Index Terms—Pneumothorax, Reinforcement Learning, Radiology Report Generation, VLMs

I. INTRODUCTION

Ensuring both accuracy and interpretability in the generation of diagnostic reports is fundamental to reliable clinical decision-making. However, existing automated solutions prioritize multi-pathology classification and fail to emulate the reasoning patterns typically employed by radiologists. Pneumothorax, a clinically urgent condition characterized by partial or complete lung collapse due to air accumulation in the pleural cavity between the lung and the chest wall [1], serves as a representative scenario in which precise localization and interpretation are directly related to patient outcomes. Recent advances in Vision-Language Models (VLMs) have enabled

joint learning of textual semantics from radiology reports and visual representations from medical images, facilitating automated report generation in clinical contexts [2]. Nevertheless, chest X-ray imaging presents unique challenges compared to natural images, including low contrast, subtle intensity differences, and overlapping anatomical structures, which significantly complicate accurate model inference [3]. Conventional Supervised Fine-Tuning (SFT) frameworks optimize model parameters by minimizing cross-entropy loss with respect to fixed reference reports. While this paradigm is effective for reproducing training data distributions, its reliance on static teacher-forcing limits the model’s capacity to maintain structural consistency and diagnostic reliability during inference. Consequently, a mismatch arises between the training objective and real-world performance, often leading to outputs that are fluent in language yet insufficiently trustworthy from a clinical perspective.

To tackle these difficulties in producing precise and interpretable pneumothorax reports, we adopt a RL-driven and segmentation-guided methodology [4] that incorporates reinforcement learning into the fine-tuning of VLMs, while also embedding pneumothorax segmentation masks into chest radiographs. The goal of this work is to replicate the real-world clinical diagnostic process for pneumothorax, which generally proceeds through three successive phases: initial medical image acquisition, followed by radiologist assessment with lesion delineation, and ultimately diagnostic report generation. By embedding this workflow into our modeling framework, we allow VLMs to more closely align with the authentic reasoning pipeline followed by radiologists.

In this work, we utilized the CANDID-PTX dataset, in particular we employed both the chest X-ray images and their corresponding segmentation masks as visual input. These inputs are used to fine-tune Qwen-VL series model under a RL-driven fine-tuning setup. Model performance is rigorously evaluated across six established metrics to ensure both diagnostic accuracy and linguistic quality. The key contribution of this study is as following:

- We propose a RL-driven fine-tuning strategy to improve

the fine-tuning performance of VLMs.

- We employ a segmentation-guided approach for pneumothorax diagnosis generation, then analyze the generalized report with zero-shot, with/without prior segmentation knowledge to validate our approach enhances both accuracy and interpretability of report generation.
- We utilize the activation heatmap to compare the focus of VLMs during the fine-tuning process.

II. RELATED WORK

Recent advances in Large Language Models (LLMs) have demonstrated that reinforcement learning can effectively align model behaviors with human preferences. Reinforcement Learning from Human Feedback (RLHF) has become a standard post-training paradigm, typically involving SFT, reward modeling, and policy optimization via algorithms such as Proximal Policy Optimization (PPO). Inspired by its success, reinforcement learning has been extended to VLMs to address cross-modal hallucination. Sun *et al.* [5] introduced Factually Augmented RLHF, leveraging human preference annotations to train a reward model that guides policy optimization toward visually grounded responses. Yu *et al.* [6] proposed RLHF-V, which incorporates fine-grained segment-level correctional feedback to provide more precise supervision beyond pairwise preference comparisons. To reduce reliance on costly human annotations, Wang *et al.* [7] proposed RL-VLM-F, where vision-language foundation models themselves serve as the feedback source for constructing reward signals.

III. MATERIALS AND METHODS

A. Dataset

In this study, our experiments are implemented on the Chest X-ray Anonymized Dataset in Dunedin - Pneumothorax (CANDID-PTX), a private dataset for pneumothorax representing 19,237 chest radiographs [8]. The age range is 16-101 years ($\mu = 60.1$ and $\sigma = 20.1$). Among the whole dataset, 3,196 chest radiography represent pneumothorax cases, while the remaining images are normal cases, with the resolution of 1024×1024 pixels in DICOM format. Under a controlled evaluation protocol, we assess the fine-tuned VLM on a randomly sampled subset of 1,000 pneumothorax-positive cases from the dataset, each containing detailed pneumothorax statement. Among these, 800 cases are utilized for model fine-tuning, 100 cases for validation, and the final 100 cases are held out as an independent test set.

The segmentation mask indicate pneumothorx region is visually combined with the original X-ray image via a semi-transparent overlay, with the transparency value configured at 0.4. The identified lesion regions are rendered using semi-transparent color highlighting, which retains the underlying anatomical details while drawing attention to clinically relevant areas. Examples are illustrated in Fig. 1.

B. Alignment via Group Relative Policy Optimization (GRPO)

For automated pneumothorax radiology report generation, we adopt Group Relative Policy Optimization (GRPO) to fine-tune the VLM via reinforcement learning. As illustrated in

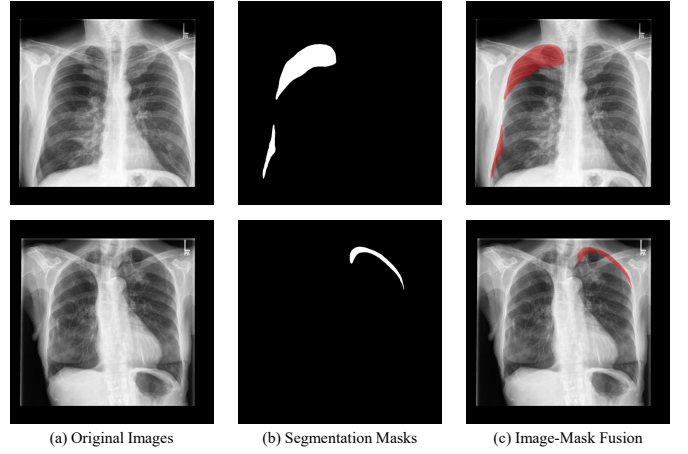


Fig. 1. Examples of image-mask fusion from CANDID-PTX dataset.

the overall data pipeline (Fig. 2), GRPO eliminates the need for a separate critic network by estimating relative advantages within a group of candidate outputs generated for the same visual input [9]. This is particularly advantageous for medical report generation, where multiple valid report structures can describe identical radiographic findings with varying degrees of diagnostic precision.

Given a chest X-ray image x and its corresponding segmentation mask m , we first perform a fusion operation to obtain a joint visual representation:

$$\tilde{x} = f_{\text{fusion}}(x, m). \quad (1)$$

1) *Notations and Candidate Generation:* Conditioned on $\tilde{x}$, the policy π_{θ} samples a candidate group of K output completion sequences $\{o_1, o_2, \dots, o_K\}$, where each completion $o_i \sim \pi_{\theta}(\cdot | \tilde{x})$ encompasses both a reasoning trace r_i (enclosed within $\langle \text{REASONING} \rangle$) and the final predicted diagnostic report $\hat{y}_i$ (enclosed within $\langle \text{SOLUTION} \rangle$), i.e., $o_i = (r_i, \hat{y}_i)$. Let y^* denote the ground-truth reference report.

Each generated completion o_i is evaluated by a task-specific scalar reward $R(\tilde{x}, o_i, y^*)$. The relative advantage A_i for candidate o_i is computed by standardizing the rewards within the sampled group:

$$A_i = \frac{R(\tilde{x}, o_i, y^*) - \mu_R}{\sigma_R}, \quad (2)$$

where μ_R and σ_R denote the mean and standard deviation of rewards across the K candidate outputs.

2) *Optimization Objective:* To stabilize training and prevent policy drift away from the base reference model π_{ref} , we optimize the policy using a clipped surrogate objective augmented with an explicit Kullback-Leibler (KL) divergence penalty:

$$L(\theta) = -\frac{1}{K} \sum_{i=1}^K L_i^{\text{clip}}(\theta) + \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}), \quad (3)$$

where $L_i^{\text{clip}}(\theta)$ represents the clipped surrogate objective for the i -th completion:

$$L_i^{\text{clip}}(\theta) = \min(r_i(\theta)A_i, \text{clip}(r_i(\theta), 1 - \epsilon, 1 + \epsilon)A_i), \quad (4)$$

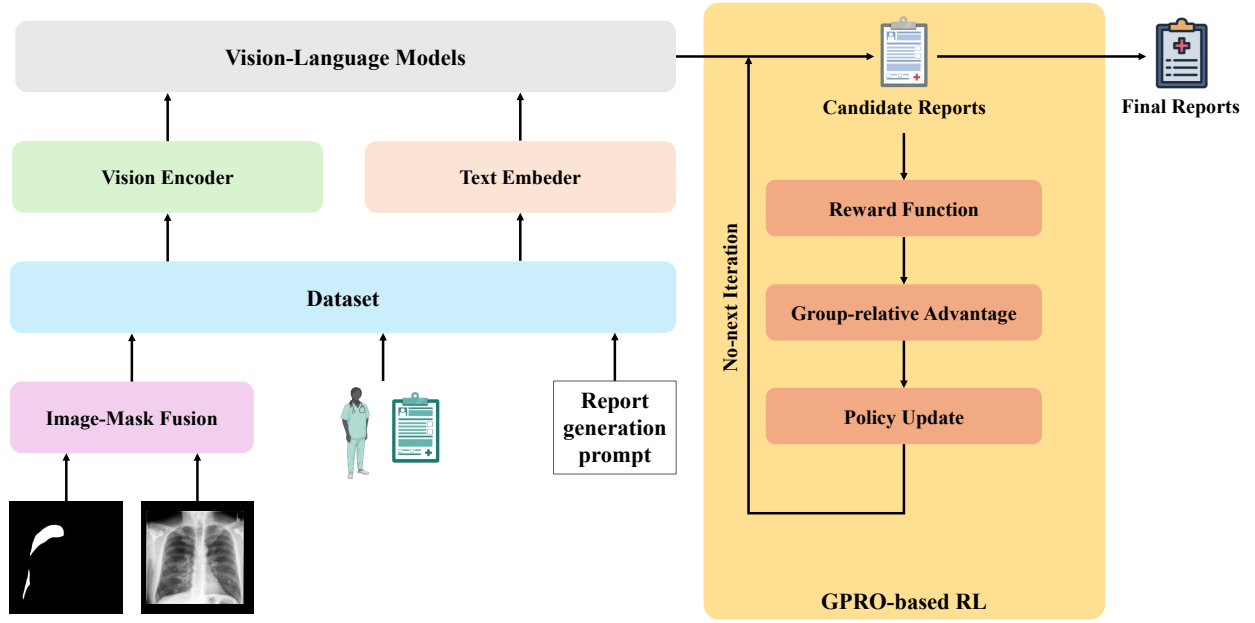


Fig. 2. Overview of pipeline for GRPO-guided radiology report generation.

where ϵ is the clipping hyperparameter, β controls the strength of the KL divergence penalty, and

$$r_i(\theta) = \frac{\pi_{\theta}(o_i | \tilde{x})}{\pi_{\theta_{old}}(o_i | \tilde{x})} \quad (5)$$

is the probability ratio between the current and previous policies.

C. Reward Functions

To enforce both structural compliance and diagnostic precision, we design a hierarchical reward function $R_{total}(o, y^*) \in [0, 2.5]$, decomposed into a Formatting Reward $R_{fmt}(o)$ and a Content Correctness Reward $R_{corr}(\hat{y}, y^*)$:

$$R_{total}(o, y^*) = R_{fmt}(o) + R_{corr}(\hat{y}, y^*). \quad (6)$$

1) *Formatting Reward (R_{fmt})*: The formatting component $R_{fmt}(o) \in [0, 0.5]$ enforces structural adherence to the chain-of-thought template. Because instruction-tuned models master tag structures quickly, we cap this reward to keep optimization focused on clinical quality:

$$R_{fmt}(o) = \max\left(0, +0.5 \cdot \mathbf{1}[\text{struct_ok}(o)] - 0.5 \cdot \mathbf{1}[\text{verbose}(o)] - 0.5 \cdot \mathbf{1}[\text{repeat}(o)]\right). \quad (7)$$

A base score of +0.5 is awarded when `<REASONING>` and `<SOLUTION>` tags appear properly and non-emptily, and the solution contains core medical entities. Penalties of -0.5 are applied for high newline-to-character ratios (> 0.25) or repetitive n -gram hallucinations.

2) *Content Correctness Reward (R_{corr})*: The primary optimization driver is the Content Correctness Reward $R_{corr}(\hat{y}, y^*) \in [0, 2.0]$, evaluated on the extracted report $\hat{y}$ against ground truth y^* . We first aggregate three specialized sub-metrics into a composite correctness score $S_{combined} \in [0, 1]$:

$$S_{combined} = 0.65S_{loc} + 0.25S_{rouge} + 0.10S_{adapt}. \quad (8)$$

To give clinical correctness four times the magnitude of structural formatting ($R_{fmt} \in [0, 0.5]$), $S_{combined}$ is scaled by a factor of 2 and capped at 2.0:

$$R_{corr}(\hat{y}, y^*) = \min(2 \cdot S_{combined}, 2.0). \quad (9)$$

The composite score integrates three clinically targeted sub-metrics:

- **Localization Score (S_{loc} , weight 0.65)**: Evaluates anatomical localization accuracy guided by the visual segmentation mask:

$$S_{loc} = 0.3 S_{diag} + 0.5 S_{lat} + 0.2 S_{zone}, \quad (10)$$

where $S_{diag} \in \{0, 1\}$ checks for core diagnostic terms (e.g., *pneumothorax*); S_{lat} checks laterality match across `{left, right, bilateral}` (1.0 for exact match, 0.2 for omission, 0.0 for conflict, and 0.5 for uninformative reference); and S_{zone} measures recall across anatomical zones (defaulting to 1.0 if none specified in reference).

- **ROUGE-L Score (S_{rouge} , weight 0.25)**: Measures phrase fluency and structural coherence using the Longest Common Subsequence (LCS) F1-score to prevent keyphrase-stuffing artifacts:

$$S_{rouge} = \frac{2 \cdot \text{LCS}(\hat{y}, y^*)}{|\hat{y}| + |y^*|}. \quad (11)$$

- **Adaptive Clinical Content Score** (S_{adapt} , **weight 0.10**): Evaluates optional clinical details without penalizing concise reports:

$$S_{\text{adapt}} = \begin{cases} \frac{\sum_{g \in \mathcal{G}_{\text{app}}} w_g \cdot \mathbf{1}[\text{hit}_g(\hat{y})]}{\sum_{g \in \mathcal{G}_{\text{app}}} w_g} & \text{if } \mathcal{G}_{\text{app}} \neq \emptyset \\ 0 & \text{otherwise} \end{cases}, \quad (12)$$

where $\mathcal{G}_{\text{app}}$ is the set of applicable term groups present in reference y^* .

D. Prompt

The prompt for pneumothorax diagnostic report generation with/without segmentation guidance is shown correspondingly in Prompt 1 and 2. The difference is marked by **bold**.

- **Prompt 1:** Consider that you are a professional radiologist with extensive experience and now you are treating a patient. Analyze the provided chest X-ray image and produce a pneumothorax diagnostic report. **In the provided image, the overlapped red area is pneumothorax.** Provide your detailed reasoning between {REASONING_START} and {REASONING_END}, and then provide your final pneumothorax diagnostic report between {SOLUTION_START} and {SOLUTION_END}.
- **Prompt 2:** Consider that you are a professional radiologist with extensive experience and now you are treating a patient. Analyze the provided chest X-ray image and produce a pneumothorax diagnostic report. Provide your detailed reasoning between {REASONING_START} and {REASONING_END}, and then provide your final pneumothorax diagnostic report between {SOLUTION_START} and {SOLUTION_END}.

E. Fine-tuned Models

Qwen-VL is a series of VLMs designed by Alibaba Cloud [10]–[12]. They introduce Naive Dynamic Resolution for adaptive visual tokenization, which enables the model to flexibly map the image of different resolutions into a dynamic number of visual tokens, and Multimodal Rotary Position Embedding (M-ROPE), which decomposes positional encoding into textual, visual and video representation. Gemma is a family of lightweight, state-of-the-art open models developed by Google DeepMind [13]. MedGemma is a collection of open-source vision-language models (VLMs) built upon variants of Gemma [14]. Leveraging MedSigLIP, a medically specialized SigLIP vision encoder, these models are optimized for medical text and image understanding through fine-tuning on diverse healthcare datasets. Table I summarizes all fine-tuned models along with their corresponding configurations and training settings.

IV. RESULTS

A. Results Before/After Fine-Tuning

As illustrated in Fig. 3, a qualitative comparison across the three representative cases reveals notable differences in

TABLE I
VLMs FOR RL-DRIVEN FINE-TUNING.

Model	Size (B)	Abbreviation
Qwen2-VL-7B-Instruct	7	QW2-07
Qwen2.5-VL-7B-Instruct	7	QW2.5-07
Qwen3-VL-8B-Instruct	8	QW3-08
Qwen3.5-4B	4	QW3.5-04
Gemma-4-E4B-IT	E4	GM4-E4
MedGemma-1.5-4B-IT	4	MedGM1.5-04

diagnostic accuracy between the SFT-fine-tuned and RL-driven fine-tuned models. The SFT-fine-tuned model tends to misidentify the laterality and severity of pneumothorax, producing descriptions inconsistent with the ground-truth clinical reports. The RL-fine-tuned model demonstrates substantially improved alignment with the original reports, correctly capturing both the affected side and the approximate size of the lesion in most cases. This suggests that RL-based optimization encourages the model to develop a more clinically grounded reasoning process rather than superficial pattern matching.

The evaluation metrics across both zero-shot inference and reinforcement learning-driven fine-tuning reveal clear, consistent advantages when integrating visual segmentation guidance. In the zero-shot setting, introducing segmentation mask overlays yields widespread performance gains across virtually all evaluated models, most notably in lateral localization accuracy, finding precision, and clinical metric scores. Diverse architectures exhibit substantial improvements in spatial alignment and anatomical grounding, confirming that visual guidance provides essential contextual cues for report generation even in the absence of task-specific fine-tuning. Following reinforcement learning-driven fine-tuning, models equipped with segmentation guidance maintain a decisive superiority over their unguided counterparts. Visual guidance consistently enhances lexical overlap, semantic coherence, clinical report completeness, and overall diagnostic correctness across the majority of evaluated backbones. Although minor trade-offs appear in isolated metrics for specific model architectures, the overall findings confirm that segmentation-guided visual input acts as a powerful structural anchor, synergizing effectively with post-training optimization to yield substantially more accurate, clinically faithful, and interpretable pneumothorax reports.

B. GRPO-Based Reasoning Analysis

A representative example of the reasoning process is shown in Figure 4. The GRPO-fine-tuned model first establishes the image orientation from the “L PA” marker and correctly maps the image-side anatomy to the patient’s anatomical laterality. It then identifies the key imaging evidence within the highlighted region, including an abrupt peripheral termination of lung markings and a distinct pleural line along the lateral chest wall. The model explicitly links these findings to the presence of pleural air and pneumothorax rather than relying solely on a direct image-to-label prediction. It further evaluates the surrounding anatomy and the implanted cardiac pacemaker as

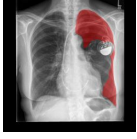
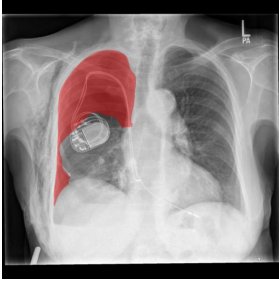
	Small left apical pneumothorax (max vertical distance 1.5 cm). Otherwise clear.	Small right apical pneumothorax is seen.	A small to moderate pneumothorax is identified in the left upper thoracic cavity.
	There is a large right pneumothorax.	There is a small right apical pneumothorax but no evidence of tension.	Chest X-ray demonstrates a large -sized pneumothorax on the patient's right side.
	There has been a left pneumothorax with loss of volume of the left lung.	Left-sided pneumothorax is seen.	A moderate-to-large pneumothorax is identified in the left lung.
(a) Original Images	(c) Pneumothorax Descriptions in Original Reports	(c) Pneumothorax Descriptions in Reports Generated by SFT-Fine-Tuned Models	(d) Pneumothorax Descriptions in Reports Generated by RL-Fine-Tuned Models

Fig. 3. Comparison of (b) original report, (c) reports generated by SFT-fine-tuned models, and (d) reports generated by RL-fine-tuned models. Red indicates descriptions consistent with the reference reports, blue denotes incorrect or contradictory statements, and green highlights clinically accurate findings that are not explicitly mentioned in the original reports.



- Image orientation:** The marker “L PA” indicates this is a Left Posteroanterior (PA) chest X-ray. Therefore, the patient’s left side is on the viewer’s right, and the right side is on the viewer’s left.
- Pneumothorax identification:** A pneumothorax is defined as air in the pleural space causing partial or complete collapse of the lung. On a PA chest X-ray, it appears as a sharp, thin line (pleural line) separating the collapsed lung from the mediastinum and diaphragm, often accompanied by visible lung parenchyma retraction or absence of lung markings beyond that line.
- Visual analysis:** The red overlay highlights an area on the left hemithorax, which corresponds to the patient’s right side. Within this region, there is a clear demarcation where lung tissue abruptly ends - no lung markings extend into the peripheral zone adjacent to the visceral pleura. There is also a visible pleural line along the lateral chest wall, consistent with a pneumothorax. The presence of a cardiac pacemaker device does not obscure the visualization of the pleural line; rather, its position confirms the anatomical location for assessing the right-sided pneumothorax. No other abnormalities such as effusion, consolidation, or mass lesions are evident in the same region.
- Clinical correlation:** While the patient has a pacemaker, this does not preclude pneumothorax diagnosis. Pacemakers can be implanted near the apex of the heart but do not typically cause pneumothorax unless there was trauma during implantation or subsequent complications.
- Conclusion:** Based on the visual evidence — the abrupt cutoff of lung markings, the presence of a pleural line, and the anatomical localization — this represents a pneumothorax on the patient’s right side (left side of the image).

Fig. 4. Example of the reasoning process generated by the GRPO-fine-tuned model, illustrating how image-derived radiographic features are progressively identified and integrated to arrive at the final pneumothorax diagnosis.

a potential confounding factor, determining that the device does not interfere with visualization of the pleural abnormality. By integrating the radiographic findings with the established anatomical orientation, the model ultimately localizes the lesion to the patient’s right hemithorax and produces the corresponding pneumothorax diagnosis. This case demonstrates a coherent progression from visual evidence extraction and anatomical reasoning to lesion localization and final report generation.

V. CONCLUSION

In this work, we investigated GRPO-based reinforcement learning fine-tuning for pneumothorax detection and localization in chest radiographs. Experimental results demonstrate

that GRPO substantially improves diagnostic accuracy across generated reports. These improvements are reflected not only in conventional text-overlap metrics, but also in clinically relevant evaluation metrics, indicating that GRPO enhances both the overall quality and clinical fidelity of the generated diagnostic reports.

We further analyzed the reasoning traces generated by the GRPO-trained model to investigate how visual evidence is translated into the final diagnostic decision. The results demonstrate that the model can identify and integrate radiographic characteristics associated with pneumothorax and subsequently use these visual cues to derive the final diagnosis and report. This reasoning process provides a more transparent connection between image-level evidence, intermediate

TABLE II
EVALUATION METRICS COMPARISON BETWEEN VLMS FOR INFERENCE TASK. **Bold** INDICATES THE HIGHER VALUE IN WITH/WITHOUT SEGMENTATION-GUIDANCE.

Model	Guidance	Lat-Acc	Lat-BAcc	Find-P	ROUGE-2	ROUGE-L	METEOR	RG-Complete	CheXbert-F1
QW2-07	N/A	0.3500	0.2482	0.0000	0.0093	0.0843	0.0119	0.0120	0.4665
	+Seg	0.4500	0.3475	0.0909	0.0158	0.1291	0.0367	0.0395	0.6855
QW2.5-07	N/A	0.2100	0.1814	0.0244	0.0175	0.1380	0.0584	0.0377	0.5976
	+Seg	0.4400	0.4354	0.0833	0.0203	0.1500	0.0577	0.0552	0.6962
QW3.5-04	N/A	0.2800	0.1723	0.1609	0.0480	0.1642	0.1181	0.0702	0.4039
	+Seg	0.9900	0.9925	0.2412	0.0601	0.1960	0.1098	0.0981	0.6963
MedGM1.5-04	N/A	0.1700	0.1500	0.1151	0.0102	0.0486	0.0222	0.0246	0.2461
	+Seg	0.9300	0.9470	0.2090	0.0566	0.1723	0.0731	0.0905	0.6514
GM4-E4	N/A	0.3300	0.2208	0.0608	0.0281	0.1357	0.0628	0.0367	0.5928
	+Seg	0.8900	0.5333	0.1862	0.0393	0.1758	0.0676	0.0693	0.6175

TABLE III
EVALUATION METRICS COMPARISON BETWEEN MODEL AFTER RL-DRIVEN FINE-TUNING. **Bold** INDICATES THE HIGHER VALUE IN WITH/WITHOUT SEGMENTATION-GUIDANCE.

Model	Guidance	Lat-Acc	Lat-BAcc	Find-P	ROUGE-2	ROUGE-L	METEOR	RG-Complete	CheXbert-F1
QW2-07	N/A	0.4400	0.3121	0.2751	0.0000	0.1938	0.0966	0.0012	0.6203
	+Seg	0.9600	0.6198	0.1010	0.0713	0.1910	0.1666	0.0333	0.5656
QW2.5-07	N/A	0.2000	0.2121	0.0588	0.0198	0.1357	0.0587	0.0394	0.6053
	+Seg	0.1100	0.3956	0.4444	0.0361	0.1465	0.1436	0.0254	0.7509
QW3.5-04	N/A	0.7300	0.6965	0.1498	0.0610	0.1993	0.1428	0.0568	0.5859
	+Seg	0.9100	0.9007	0.3750	0.0630	0.1986	0.1135	0.0811	0.6461
MedGM1.5-04	N/A	0.7100	0.5095	0.1532	0.0423	0.1434	0.0423	0.0562	0.6490
	+Seg	0.9800	0.9841	0.2366	0.0569	0.1750	0.0538	0.0852	0.7524
GM4-E4	N/A	0.4100	0.2988	0.0811	0.0326	0.1521	0.0738	0.0334	0.6395
	+Seg	0.9000	0.5595	0.2021	0.0398	0.1828	0.0743	0.0643	0.6207

reasoning, and the final diagnostic output, offering an interpretable view of the model’s decision-making process.

Finally, we conducted a controlled comparison between models with and without mask-based guidance. The results demonstrate that the proposed guidance mechanism significantly improves pneumothorax localization, showing that explicit spatial guidance can effectively direct the model toward lesion-relevant regions and improve its ability to identify the anatomical location of pneumothorax. Overall, our findings demonstrate that GRPO-based optimization, together with spatial mask guidance, provides an effective framework for improving the accuracy, clinical relevance, interpretability, and localization capability of multimodal models for pneumothorax diagnosis.

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